RiskAdaptive: An Agentic AI Framework for Risk-Aware Intrusion Detection and Security Prioritization in Edge-IoT Networks

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***Abstract*—The rapid expansion of Edge-IoT deployments has introduced complex security challenges due to device heterogeneity, distributed architectures, limited computational resources, and increasing cyber attack sophistication. Traditional intrusion detection systems primarily focus on binary attack classification and lack contextual risk prioritization required for automated incident response. This paper presents RiskAdaptive, an agentic AI-based framework that integrates intrusion detection with dynamic risk assessment for Edge-IoT security environments. The proposed framework combines machine learning based threat identification with a risk-aware decision layer that evaluates attack severity, asset impact, and contextual threat factors. The system is evaluated using the `TON-IoT` network telemetry dataset and compared against conventional machine learning baselines including Logistic Regression and Random Forest. Experimental results demonstrate that RiskAdaptive achieves 99.27% accuracy, 99.54% F1-score, and 99.94% precision while providing additional risk prioritization capabilities beyond conventional IDS classification. Attack-specific analysis demonstrates effective identification of major threat categories including ransomware, backdoor, and man-in-the-middle attacks. The proposed approach establishes a foundation for autonomous, explainable, and risk-aware security operations in distributed Edge-IoT environments.**

***Index Terms*—Agentic AI, Edge-IoT Security, Intrusion Detection System, Risk Assessment, `TON-IoT` Dataset, Machine Learning, Cybersecurity Automation**

I. Introduction

The rapid adoption of Internet of Things (IoT) technologies has transformed modern computing environments by enabling interconnected systems across industrial automation, healthcare, smart cities, and critical infrastructure. However, the expansion of IoT deployments towards distributed Edge-IoT architectures has significantly increased the cybersecurity attack surface. Unlike conventional enterprise networks, Edge-IoT environments consist of heterogeneous devices with limited computational capabilities, dynamic communication patterns, and decentralized deployment characteristics.

Traditional intrusion detection systems (IDS) primarily focus on identifying whether network activity is malicious or benign. Although machine learning based IDS approaches have demonstrated strong classification performance, they generally provide limited contextual understanding regarding attack severity, asset importance, and operational impact. In practical Security Operations Center (SOC) environments, analysts require not only detection capability but also risk prioritization to determine which alerts require immediate response.

Recent research has explored artificial intelligence techniques for enhancing IoT security. Dataset-driven approaches such as TON-IoT have enabled standardized evaluation of intrusion detection models across heterogeneous IoT and industrial environments. Machine learning models including Random Forest, Support Vector Machines, and deep learning architectures have achieved high detection performance; however, these approaches commonly operate as isolated classifiers without integrated risk reasoning or response prioritization.

Agentic AI introduces a different perspective by enabling systems to coordinate multiple decision-making components for autonomous security operations. In cybersecurity applications, agent-based architectures can combine threat perception, contextual analysis, risk estimation, and response planning. However, existing studies frequently focus on either intrusion detection or automated response, leaving a gap in integrating accurate detection with adaptive risk assessment for Edge-IoT environments.

This paper presents RiskAdaptive, an agentic AI framework that extends traditional intrusion detection by incorporating a risk-aware decision layer. The proposed system combines machine learning based threat identification with contextual risk scoring to classify detected events according to operational severity. Unlike conventional IDS approaches that terminate after attack classification, RiskAdaptive provides additional security intelligence for prioritizing incidents and supporting automated security decision workflows.

The major contributions of this work are summarized as follows:

- A risk-aware agentic AI framework is proposed for Edge-IoT intrusion detection and incident prioritization.
- A dynamic risk assessment layer is integrated with intrusion classification to transform binary detection outputs into severity-aware security decisions.
- Extensive evaluation is performed using the TON-IoT

network telemetry dataset with comparison against conventional machine learning baselines including Logistic Regression and Random Forest.
- Attack-specific analysis is presented to evaluate detection capability and risk distribution across different threat categories.

The remainder of this paper is organized as follows. Section II reviews related research in IoT intrusion detection, agentic AI security frameworks, and risk-aware cybersecurity approaches. Section III presents the proposed RiskAdaptive methodology. Section IV describes the experimental setup and evaluation procedure. Section V discusses experimental results and findings. Section VI concludes the paper and highlights future research directions.

II. Related Work

Existing IoT intrusion detection research has extensively explored machine learning and deep learning approaches for identifying malicious network activities. Traditional machine learning based IDS approaches have demonstrated strong classification capability; however, most existing methods primarily focus on attack detection without providing context-aware prioritization of security events.

Alsaedi et al. introduced the `TON-IoT` telemetry dataset, enabling evaluation of machine learning approaches for IoT and industrial IoT security scenarios [1]. Several studies have investigated machine learning and deep learning based intrusion detection models. Shone et al. proposed a deep learning approach for network intrusion detection using representation learning techniques [2]. Similarly, Vinayakumar et al. demonstrated the effectiveness of deep neural architectures for large-scale intrusion detection problems [3].

Ensemble learning and anomaly detection approaches have also been widely explored. Mirsky et al. proposed Kitsune, an ensemble of autoencoders designed for online network intrusion detection [4]. Although these approaches improve detection performance, they generally do not incorporate security risk estimation or automated response prioritization.

Recent research has explored explainable AI and intelligent security analytics to improve operational usefulness. Recent studies have investigated machine learning based intrusion detection with explainability mechanisms. However, existing approaches still lack an integrated framework combining detection, risk scoring, severity estimation, and decision support.

TABLE I
COMPARISON OF RISKADAPTIVE WITH EXISTING APPROACHES

Approach	**Detection**	**Risk Score**	**Severity**	**Response Support**
ML based IDS [5]	Yes	No	Limited	No
`TON-IoT` ML Models [1]	Yes	No	No	No
Deep Learning IDS [3]	Yes	No	No	No
Kitsune [4]	Yes	No	No	No
RiskAdaptive	Yes	Yes	Yes	Yes

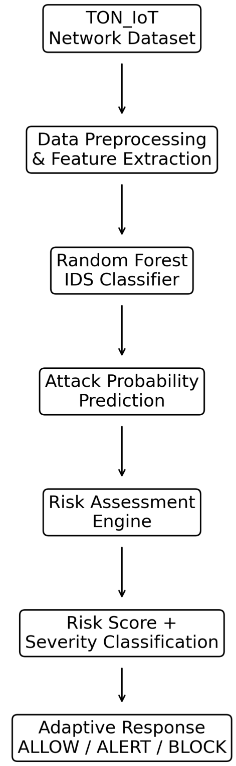

Fig. 1. Architecture of the proposed RiskAdaptive framework.

Unlike existing IDS approaches, RiskAdaptive introduces an additional risk-aware decision layer that converts detection outputs into actionable security priorities. The proposed framework therefore bridges the gap between intrusion detection and autonomous security operations by combining threat identification, risk estimation, and severity-based prioritization.

III. Proposed Methodology

A. RiskAdaptive Framework Overview

The proposed RiskAdaptive framework introduces a risk-aware intrusion detection architecture for Edge-IoT environments. Unlike traditional IDS solutions that provide only binary attack classification, the proposed framework integrates detection with contextual risk evaluation to prioritize security events based on their potential impact.

Figure 1 illustrates the overall architecture of the proposed system. The framework consists of four major components: data ingestion, intrusion detection, agentic risk analysis, and security decision support.

B. Data Processing and Feature Representation

The framework utilizes network telemetry data obtained from the `TON-IoT` dataset. The preprocessing pipeline performs data cleaning, feature transformation, and preparation of network attributes for machine learning based classification.

The processed feature representation enables the intrusion detection model to identify malicious activities while maintaining compatibility with heterogeneous IoT network environments.

C. Intrusion Detection Module

The intrusion detection component uses supervised machine learning classification to distinguish between normal and malicious network activities. A Random Forest based classifier is employed due to its ability to model nonlinear relationships among network security features.

The output of this module consists of the predicted attack category and associated confidence information, which is forwarded to the risk assessment layer.

D. Agentic Risk Assessment Engine

The core contribution of RiskAdaptive is the integration of an agentic risk evaluation layer above the intrusion detection module. The risk engine analyzes detected events using contextual security attributes including attack category, severity level, and potential impact.

Instead of treating every detected attack equally, the framework generates a continuous risk score that enables prioritization of security incidents.

E. Risk Scoring and Severity Classification

The generated risk score represents the combined impact of detected security events. Based on predefined risk thresholds, events are categorized into three severity levels:

- Low: Minimal security impact requiring monitoring.
- Medium: Suspicious activity requiring further analysis.
- High: Critical events requiring immediate attention.

This adaptive prioritization mechanism reduces alert overload and provides security operators with actionable information.

F. Decision Support Workflow

The final stage converts detection results into security insights by combining attack identification, risk estimation, and severity classification. The framework therefore extends conventional IDS systems from attack detection toward intelligent security prioritization.

IV. Experimental Setup

A. Dataset Description

The proposed RiskAdaptive framework is evaluated using the `TON-IoT` network telemetry dataset, which contains diverse IoT network traffic samples representing both normal and malicious activities. The dataset includes multiple attack categories, enabling evaluation of the framework under different threat scenarios.

The experiments consider attack detection capability as well as risk-based prioritization performance across different attack types.

B. Experimental Configuration

The dataset is processed through preprocessing and feature transformation stages before model training and evaluation. The intrusion detection model is trained using labeled network traffic samples, while the risk assessment module operates on the generated detection outputs.

The evaluation pipeline consists of three stages:

1) Network traffic preprocessing and feature preparation.
2) Attack classification using machine learning based IDS.
3) Risk scoring and severity assignment using the RiskAdaptive engine.

C. Baseline Comparison

To evaluate the effectiveness of the proposed approach, comparisons are performed against conventional machine learning based IDS models. The baseline models include Logistic Regression and Random Forest classifiers.

The comparison focuses on standard classification metrics including accuracy, precision, recall, F1-score, false positive rate (FPR), and false negative rate (FNR).

D. Evaluation Metrics

The following metrics are used for performance evaluation:

- Accuracy: Measures the overall correctness of predictions.
- Precision: Represents the proportion of correctly identified attacks among predicted attacks.
- Recall: Measures the ability to detect actual attack instances.
- F1-score: Represents the harmonic mean of precision and recall.
- False Positive Rate: Measures incorrectly generated alerts.
- False Negative Rate: Measures missed attack instances.

E. Attack Scenario Evaluation

In addition to overall classification performance, attack-specific analysis is conducted for different threat categories including backdoor, ransomware, and man-in-the-middle attacks.

This evaluation demonstrates the capability of RiskAdaptive to assign different risk levels according to attack characteristics rather than treating all detected events uniformly.

V. Results and Discussion

A. Overall Performance Evaluation

The performance of RiskAdaptive was evaluated against conventional machine learning baselines using the `TON-IoT` network telemetry dataset. Logistic Regression and Random Forest models were considered as baseline intrusion detection approaches.

TABLE II
PERFORMANCE COMPARISON WITH BASELINE MODELS

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	98.87%	99.30%	99.25%	99.28%
Random Forest	99.27%	99.94%	99.15%	99.54%
RiskAdaptive	99.27%	99.94%	99.15%	99.54%

The results demonstrate that RiskAdaptive maintains competitive classification performance while extending conventional IDS capability through risk-aware analysis. Unlike traditional classifiers that produce only attack labels, the proposed framework additionally generates security priorities based on estimated risk and severity.

B. Attack-Specific Analysis

To evaluate robustness across different threat categories, the proposed framework was analyzed against multiple attack scenarios including backdoor, ransomware, and man-in-the-middle attacks.

Backdoor attacks achieved high detection capability with elevated risk scores, indicating effective identification of severe attack patterns. Ransomware traffic was also successfully detected with high confidence, demonstrating the capability of RiskAdaptive to prioritize impactful security events.

Man-in-the-middle attacks showed comparatively lower detection performance due to their stealth characteristics and traffic similarity with legitimate communication patterns. This highlights the challenge of detecting low-visibility attacks in IoT environments.

C. Risk Prioritization Capability

Beyond detection accuracy, RiskAdaptive provides an additional decision-support layer by categorizing detected events according to severity levels. The framework separates security events into HIGH, MEDIUM, and LOW risk categories, enabling analysts to prioritize critical incidents.

This capability addresses a limitation of conventional IDS systems, where every detected attack is generally treated with equal priority. By incorporating contextual risk scoring, RiskAdaptive improves the practical usability of automated security monitoring.

D. Discussion

The experimental findings indicate that combining machine learning based intrusion detection with agentic risk analysis provides a more complete security workflow. While baseline models demonstrate strong classification performance, they lack mechanisms for contextual decision making.

RiskAdaptive extends existing IDS approaches by transforming detection outputs into actionable security intelligence. The framework therefore provides a bridge between automated attack detection and intelligent incident response for Edge-IoT environments.

VI. CONCLUSION

A. Conclusion

This paper presented RiskAdaptive, a risk-aware intrusion detection framework designed for IoT security environments. Unlike conventional IDS approaches that primarily focus on attack classification, the proposed framework integrates intrusion detection with contextual risk assessment and severity prioritization.

The experimental evaluation on the `TON-IoT` dataset demonstrates that RiskAdaptive achieves competitive detection performance while providing additional security insights through risk scoring. The framework successfully differentiates security events according to their potential impact, enabling more effective alert prioritization.

The findings indicate that combining machine learning based detection with intelligent risk analysis can improve the practical usability of IoT security monitoring systems. Future work will focus on extending the framework with adaptive learning strategies, real-time deployment evaluation, and integration with larger heterogeneous IoT environments.

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